

Mu-Ko (Claire) Chou

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RESEARCH INTERESTS

Mechanistic Interpretability; Multilingual Legal NLP

EDUCATION

National Cheng Kung University, Tainan, Taiwan Feb. 2025 – Expected Feb. 2027
M.S. in Intelligent Technology Systems, Miin Wu School of Computing GPA: 4.19/4.30
Advisor: Prof. Shao-Man Lee

National Cheng Kung University, Tainan, Taiwan Sep. 2019 – Jun. 2024
B.A. in Political Science, College of Social Sciences

Technical University of Darmstadt, Darmstadt, Germany Spring 2023
Exchange Student

RESEARCH EXPERIENCE

National Cheng Kung University, Tainan, Taiwan
SocioLegal Lab, Miin Wu School of Computing Advisor: Prof. Shao-Man Lee

M.S. Thesis Research Feb. 2026 – Present

Research: *Mechanistic Interpretability of Multilingual Legal Concepts*

- Investigate whether multilingual large language models encode shared representations of constitutional concepts using sparse autoencoders (SAEs).
- Developed a cross-lingual analysis pipeline over 8,654 constitutional judgments in 25 languages, identifying concept-associated SAE features and tracing their activations back to source passages.
- Resulted in the preprint *Tracing Constitutional Concepts Across Languages with Sparse Autoencoder Features*.

Part-Time Research Assistant

Research: *Retrieval-Augmented Generation and Evaluation for Taiwanese Legal Documents* Aug. 2025 – Present

- Develop TaDELS-RAG, a retrieval-augmented generation system and evaluation benchmark for source-grounded question answering over Taiwanese legal documents.
- Co-developed a benchmark of 211 legal documents and 648 human-reviewed question-answer pairs, together with an entity-oriented evaluation framework for answer faithfulness and hallucination.
- Resulted in the co-first-authored *TaDELS-RAG: Entity-Oriented Evaluation Dataset for Retrieval-Augmented Generation in Taiwanese Law*, presented at the 2026 Empirical Legal Studies Workshop.

Research: *Intersectional and Cross-Cultural Biases in Large Language Models* 2024 – 2025

- Investigated intersectional and cross-cultural biases in LLM moral decision-making across demographic and cultural profiles.
- Evaluated 5 LLMs across 1,769 demographic profiles, 9 cultural clusters, and 38,350 Moral Machine decisions using Average Marginal Component Effects (AMCE) and human-model distance measures.
- Analyzed interaction effects across demographic attributes; findings were presented by Prof. Shao-Man Lee at the Young Scholar Workshop on Empirical Legal Studies in the Sinophone Region (2025).

Full-Time Research Assistant Jul. 2024 – Jan. 2025

Government Project: *AI-Assisted Prosecution System, Ministry of Justice (Taiwan)*

- Served as Technical Project Manager for a government legal-AI project, coordinating requirements, implementation, testing, and communication between prosecutors, government stakeholders, and the technical team.
- Contributed hands-on to NLP-based processing of investigative records and AI-assisted document workflows, including extraction and validation of key case information from police and interrogation records.

PUBLICATIONS & PRESENTATIONS

Manuscripts Under Review

Tracing Constitutional Concepts Across Languages with Sparse Autoencoder Features.

Shao-Man Lee, **Mu-Ko Chou**

In Preprint, 2026

Workshop Papers & Presentations

TaDELS-RAG: An Entity-Oriented Evaluation Dataset for Retrieval-Augmented Generation in Taiwanese Law.

Shao-Man Lee*, **Mu-Ko Chou***, Chun-Che Huang, Chin-You Hou

Empirical Legal Studies Workshop 2026, Aug. 5–6, 2026.

*Equal contribution.

Navigating Judicial Overload: A Multi-Agent System for Argument Tracking in Constitutional Courts.

Shao-Man Lee, **Mu-Ko Chou**, Yu-Chiao Chen, Chun-Che Huang, Chun-Chieh Niu, Chin-You Hou

Computational Legal Studies Workshop 2025, Singapore Management University, Sep. 8–9, 2025.

INDUSTRY EXPERIENCE

Cathay Financial Holdings, Kaohsiung, Taiwan

Cathay Development Center (CDC)

Feb. 2024 – Jun. 2024

Local Community Engagement Intern

- Developed and evaluated a GCP-based RAG system for financial regulatory knowledge retrieval, comparing multiple retrieval and chaining strategies with context precision, context recall, and answer relevancy.
- Implemented PostgreSQL data structures and source-attribution mechanisms to improve response traceability and reduce hallucinations.

HONORS AND AWARDS

The Phi Tau Phi Scholastic Honor Society, Honorary Member

2026

Outstanding Student Scholarship, Miin Wu School of Computing, NCKU

2025 – 2026

Competitive scholarship for top-ranked master's admits; renewed for a second year based on academic and research performance.

SELECTED COURSEWORK

Natural Language Processing; Advanced Legal Large Language Model Implementation; Fundamentals of Deep Learning Networks: Theory and Industrial Applications; Singular Value Decompositions and Data Analysis; Artificial Intelligence and Democratic Governance; Computer Vision and Deep Learning

SKILLS

Programming: Python, SQL, LaTeX

ML/NLP: PyTorch, Hugging Face Transformers, Retrieval-Augmented Generation (RAG), Sparse Autoencoders (SAEs)

Tools: Git, PostgreSQL, Google Cloud Platform (GCP)